

MEB-600 Series

Combined Thermal-Resistance & E-Beam Evaporation System

Combined Process Capability

Single chamber houses both e-beam evaporation and thermal resistance evaporation

Customizable Platform

Chamber size, substrate size, source count, heating, pumping, and accessories tailorable

E-beam Source

10 kW e-type gun, 270° deflection, 6×17 cc water-cooled crucibles, auto-indexing, programmable X-Y scan

Thermal Resistance Source

Dual evaporation electrodes, 3 kW supply, automatic switching with shutter control

Closed-Loop Thickness Control

INFICON quartz-crystal monitor with auto shutter for precise film thickness

Substrate Heating

Up to 300 °C planar / 200 °C umbrella holder, PID ±1 °C control

Automated Recipe Operation

Siemens PLC + Advantech IPC, Windows GUI, full process logging and recipe replay

Safety & Diagnostics

Interlocks, water/pump/comms alarms, power-loss recipe memory



MEB-600 — baseline configuration

Specifications

Specification	Parameters
Configuration	Combined e-beam + thermal-resistance evaporation, single vacuum chamber
Chamber	SS304, vertical cylindrical, front-opening, water-cooled walls; $\Phi 600 \times H600$ mm (baseline, larger/smaller available)
Footprint / Mass	L 2200 × W 1500 × H 2000 mm; ~1100 kg
Ultimate Vacuum	6.7×10^{-5} Pa (at ≤55% RH)
Pump-down Time	Atm → 9.0×10^{-4} Pa in ≤35 min after brief N ₂ -vented opening
Pressure Rise	≤5.0 Pa after 12 h with system off
Vacuum Pumps	KYKY 1200 L/s composite turbomolecular pump + ULVAC (Ningbo) 12 L/s rotary pump; gate-valve isolated
Vacuum Gauging	Digital combined gauge — Pirani (low vacuum) + cold cathode (high vacuum)
Substrate Holder	Rotating planar holder, max $\Phi 6"$ single wafer , 1–10 rpm continuously variable; umbrella/dome holder available

Specification	Parameters
Substrate Heating	Planar: RT–300 °C (quartz lamp); Umbrella: RT–200 °C (sheathed); PID ± 1 °C
Source-to-Substrate	330–370 mm
E-Beam Gun	E-type, 270° deflection; max power 10 kW; anode –8/–10 kV (selectable); 0–1000 mA @ 10 kV
E-Beam Scan	X-Y deflection ± 2.0 A each; 10–250 Hz, programmable per recipe
Crucibles	6 × 17 cc oxygen-free copper, water-cooled, automatic indexing, automatic shutter
Thermal Resistance	Two evaporation electrode sets with auto-switching; 3 kW power supply; compatible with boats and helical filaments; auto shutters
Thickness Control	INFICON quartz-crystal monitor; closed-loop with source shutters; rate range 0–9999.9 Å/s (0.1 Å/s resolution); thickness resolution 1 Å
Uniformity	$\leq \pm 5\%$ over $\Phi 6"$ (baseline)
Control System	Siemens PLC + Advantech IPC, Windows-style GUI, mouse/keyboard; auto and manual modes; recipe programming and run logging
Safety / Alarms	Interlocks; auto power-loss memory; recipe validation; alarms for water failure, TMP fault, communication fault
Included Utilities	Recirculating chiller; silent air compressor